

Chatbot Workflow Documentation

Simple version updated 2026-07-14 11:23. This document explains the user flow and backend flow without heavy technical terms.

Purpose

The backend lets users log in, upload documents, ask questions, and receive answers based on their own files. It also saves chats so users can return later.

Full Workflow In One View

```
User logs in
-> User starts a chat
-> User uploads documents
-> Backend reads and splits the documents
-> User asks a question
-> Backend finds useful document parts
-> AI writes an answer
-> Backend saves the conversation
```

1. User Signs Up Or Logs In

- The user creates an account with a username, email, and password.
- The backend saves the password safely in a scrambled form.
- When the user logs in, the backend gives the browser a short-time login key and a longer-time login key.
- The short-time key is used for normal requests.
- The longer-time key helps the user stay logged in without entering the password again.
- When the user logs out, the longer-time key is marked as no longer valid.

2. User Starts A Chat

- The user opens a new chat.
- The backend creates a conversation record.
- Every message in that chat is saved.
- The user can return to the chat later and continue from where they stopped.
- The system can give the chat a short title after the first messages.

3. User Uploads A Document

- The user uploads a file, such as a PDF, Word document, or text file.
- The backend checks that the file type is allowed.
- The backend checks that the file is not too large.
- The file is saved on the server.

- The document is linked to the user, and it can also be linked to a specific chat.

4. Backend Reads The Document

- The backend opens the uploaded file.
- It reads the text from the file.
- Long text is split into smaller pieces.
- These smaller pieces are saved so the chatbot can use the right part later.
- This makes the chatbot faster and more focused because it does not need to read the whole document every time.

5. User Asks A Question

- The user types a question in the chat.
- The backend saves the user's message.
- The backend looks through the saved document pieces.
- It chooses the pieces that seem most useful for the question.
- The backend sends the question, recent chat history, and useful document pieces to the AI model.
- The AI writes an answer based on that information.
- The answer is sent back to the user and saved in the chat.

6. User Continues Or Manages The Chat

- The user can ask follow-up questions.
- The backend keeps the conversation history so the chat still makes sense.
- The user can view old chats.
- The user can delete chats.
- The user can export a chat.
- The user can give feedback or ask the system to generate another answer.

What Each Main Area Does

Area	Simple Purpose
Auth	Handles signup, login, logout, and staying logged in.
Users	Stores user account details and profile information.
Conversations	Creates and manages chat threads.
Messages	Saves user messages and assistant answers.
Documents	Uploads files and prepares their text for chat.
AI Service	Builds the question with useful context and asks the AI for an answer.

Simple Example

If a user uploads an employee policy document and asks, 'How many leave days do I get?', the backend finds the useful leave-policy text from the document and sends it with the question to the AI. The AI then answers using that document text.